# EVALUATION OF A 30-HOUR ONLINE COURSE ON DATABASE MANAGEMENT SYSTEMS

## A PROJECT REPORT

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Certify that this project report titled "**EVALUATION OF A 30-HOUR ONLINECOURSE ON DATABASE MANAGEMENT SYSTEMS**" is the bonafide work of "**Aditya Mukherji [RA2411030030052]**", who has completed the projectwork under my supervision. Certified further, that apparently the work detailed thusshapes no other task report or paper based on which a degree or grant was given on aprior event on this or some other candidate

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**Aditya Mukherji**

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# ABSTRACT

This report presents an in-depth evaluation of a 30-hour online course on DatabaseManagement Systems (DBMS), focusing on three critical components: video lectures,assignments, and practical sessions. The purpose of this evaluation is to determine howeffectively the course supports second-year undergraduate students in understandingfundamental DBMS concepts and applying them in real-world scenarios.

The study adopts a mixed-method approach, combining both qualitative and quantitativetechniques. The evaluator personally completed the entire course to gain first-handexperience, while a structured survey was conducted among 30–50 second-year B.Techstudents using a Likert scale ranging from 1 to 5. The collected data was analyzed usingstatistical tools to identify patterns, strengths, and weaknesses.

The findings reveal that the video lectures are well-structured and conceptually strong,providing clarity on core topics such as relational models and SQL queries. However,assignments show inconsistency in difficulty levels, and practical sessions facechallenges related to tool accessibility and implementation. Based on these observations,several recommendations are proposed to enhance the overall learning experience andimprove alignment with academic requirements.

# CHAPTER 1
# INTRODUCTION

### *1.1 Background*

Database Management Systems (DBMS) play a fundamental role in the field ofcomputer science and information technology. In today's data-driven world, organizations rely heavily on databases to store, manage, and retrieve vast amounts ofinformation efficiently. From banking systems and e-commerce platforms to healthcareand social media, DBMS is a backbone technology that ensures data integrity, security,and accessibility. For students pursuing B.Tech in Computer Science or Information Technology, DBMSis a core subject that lays the foundation for advanced topics such as data analytics, artificial intelligence, and distributed systems. However, traditional classroom teachingoften lacks sufficient practical exposure and flexibility. This gap is effectively addressedby online courses, particularly Massive Open Online Courses (MOOCs), which providehigh-quality content from reputed institutions.

### *1.2 About the Selected Course*

The selected course is a 30-hour online DBMS course available on a recognized learningplatform. It is designed to cover essential database concepts through a structuredcombination of video lectures, assignments, quizzes, and practical exercises. The courseis divided into multiple modules, each focusing on a specific aspect of DBMS, ensuringa gradual and comprehensive learning process. The course also offers certification upon successful completion, making it beneficial foracademic as well as professional development. The instructor adopts a conceptual teaching approach supported by examples and demonstrations.

### *1.3 Need for Evaluation*

With the increasing popularity of online learning, it becomes essential to evaluate theeffectiveness of such courses in achieving their intended learning outcomes. Not allcourses deliver the same level of quality, and students often face challenges related tounderstanding, engagement, and practical implementation. This evaluation is necessary to determine whether the selected DBMS course effectivelysupports second-year students in grasping core concepts and developing practical skills.It also helps identify gaps and areas for improvement.

### *1.4 Objectives*

The primary objective of this study is to evaluate the effectiveness of the selected onlineDBMS course. Specifically, it aims to analyze the quality of video lectures in terms ofclarity and engagement, assess the relevance and difficulty of assignments, and evaluatethe usefulness of practical sessions in providing hands-on experience. Additionally, thestudy aims to propose recommendations to enhance the course structure and delivery.

### *1.5 Scope and Limitations*

The scope of this study is limited to evaluating the course from a student's perspectiveusing personal experience and survey responses. While the findings provide valuableinsights, they are subject to certain limitations. The sample size of respondents isrelatively small, and the evaluation does not include inputs from instructors or coursedevelopers. Furthermore, technical issues faced during practical sessions may varydepending on individual system configurations

### *1.6 Report Structure*

This report is organized into six chapters. Chapter 1 introduces the topic and outlines theobjectives. Chapter 2 provides an overview of the course content. Chapter 3 explains themethodology used for evaluation. Chapter 4 presents the analysis and findings. Chapter5 discusses the results and provides recommendations, and Chapter 6 concludes thereport.

# CHAPTER 2
# *COURSE OVERVIEW AND DBMS CONCEPTS COVERED*

The selected 30-hour online course on Database Management Systems is systematically designed to provide a comprehensive understanding of both theoretical and practical aspects of database systems. The course is divided into 16 modules, each focusing on specific DBMS concepts and gradually building the learner's knowledge from foundational topics to advanced techniques.

## *2.1 Module-wise Breakdown*

| *Mo dule* | *Topic* | *Key Concepts* | *Duration* |
|---|---|---|---|
| 1 | Intro to DBMS | File System vs DBMS, Architecture, Schemas | 1h 45m |
| 2 | ER Modelling | Entities, Attributes, Relationships, Cardinality | 2h 10m |
| 3 | Relational Model | Tables, Keys: Primary / Foreign / Candidate | 1h |
| 4 | Relational Algebra | Select, Project, Join, Union, Difference | 50m |
| 5 | SQL -- DDL | CREATE, ALTER, DROP, TRUNCATE, Constraints | 2h |
| 6 | SQL -- DML & Queries | INSERT, UPDATE, DELETE, SELECT, GROUP BY | 1h 55m |
| 7 | SQL -- Advanced | Joins, Subqueries, Views, Indexes, Triggers | 2h 30m |
| 8 | Normalisation I | Functional Dependencies, 1NF, 2NF, 3NF | 2h |
| 9 | Normalisation II | BCNF, 4NF, Lossless Decomposition | 20m |
| 10 | Transactions | ACID Properties, Transaction States, Serializability | 1h |
| 11 | Concurrency Control | Locking, 2PL, Deadlock Detection | 45m |
| 1 | Recovery Systems | Log-Based Recovery, Checkpoints, ARIES | 2h 00m |
| 2 | Indexing | B-Trees, B+ Trees, Dense/Sparse, Hashing | 1h |
| 1 | | | 1h 50m |
| 3 | | | 40m |

| Module | Topic | Key Concepts | Duration |
|---|---|---|---|
| 14 | Query Optimisation | Cost-Based, Execution Plans, Heuristics | 1h 35m |
| 15 | NoSQL Databases | CAP Theorem, MongoDB Basics, Document Model | 1h 20m |
| 16 | IMDB Case Study | End-to-End SQL on Real-World IMDB Dataset | 1h 40m |

***Table 2.1: Module-wise Breakdown of the Scaler DBMS Course***

### *2.2 Key Topics Covered*

Thecoursecomprehensively covers all major DBMS topics required for both academic understanding and software industry preparation:

- ***ER Modelling:*** Strong and weak entities, attribute types, cardinality and participation constraints, ER-to-relational schema conversion.
- ***Relational Algebra:*** Formal query operations -- selection, projection, join, union, intersection, and difference.
- ***SQL (DDL, DML, Advanced):*** Three dedicated modules covering complete SQL syntax from basic CREATE/INSERT to complex multi-table joins, correlated subqueries, views, and triggers.
- ***Normalisation (1NF to BCNF):*** Functional dependencies, anomaly elimination, lossless decomposition, and dependency preservation.
- ***Transactions and ACID:*** Transaction lifecycle, concurrency schedules, serializability, and isolation levels.
- ***Concurrency Control:*** Lock-based protocols, two-phase locking (2PL), deadlock detection and prevention.
- ***Indexing and Query Optimisation:*** B+ Tree structures, hash indexing, cost-based query plan selection.
- ***IMDB Case Study:*** Applied SQL project on a real-world dataset -- a unique practical module not typically found in standard MOOCs.

- ***Module Challenges:*** 16 graded challenges -- one per module -- to test conceptual understanding.
- ***Progress Tracking:*** Visual completion indicators for each module, lesson, and challenge.
- ***Certificate of Excellence:*** Awarded automatically upon 100% course completion, downloadable and shareable to LinkedIn.
- ***Lifetime Access:*** No enrolment deadline; content available indefinitely for revision.
- ***IMDB Dataset:*** A downloadable real-world dataset provided for Module 16 applied SQL exercises.
- ***Discussion Forum:*** Community threads for doubt resolution and peer discussion.

# CHAPTER 3
# *METHODOLOGY*

This chapter outlines the research methodology adopted to evaluate the effectiveness of the selected 30-hour online DBMS course. The methodology is designed to ensure a systematic, unbiased, and comprehensive analysis by combining both numerical data and subjective student feedback.

## *3.1 Research Approach*

The study adopts a ***mixed-method research approach***, integrating both quantitative and qualitative techniques. The quantitative component focuses on numerical ratings collected through a structured Likert scale, enabling statistical comparison of various course parameters. The qualitative component captures descriptive comments and suggestions, providing deeper contextual understanding of student experiences. Triangulating both data streams ensures that the evaluation is data-driven as well as contextually rich.

## *3.2 Data Collection*

### *3.2.1 Personal Course Completion*

The evaluator personally completed all 16 modules and 90 lessons of the Scaler DBMS course over a four-week period, accumulating approximately 29 hours of video content. All 16 module challenges were attempted. Detailed notes were maintained on conceptual clarity, pacing, visual aid quality, assignment relevance, and practical session usability.

### *3.2.2Peer Survey*

A structured questionnaire was developed using ***Google Forms*** and distributed to 30-50 second-year B.Tech (CSE/IT) students who had completed at least 60% of the Scaler DBMS course. The survey comprised 22 items across three sections -- video lectures, assignments, and practicals -- rated on a 1-5 Likert scale, followed by two open-ended questions on strengths and suggested improvements.

## *3.3 Evaluation Parameters*

The course was analysed based on three major components, each assessed using specific parameters:

***Video Lectures***

- ***Clarity of Concepts:*** Ease of understanding core DBMS topics such as SQL joins, normalisation, and ACID properties.
- ***Pacing and Length:*** Appropriateness of lecture duration (ideal: under 20 minutes) and speaking pace.
- ***Visual Aids:*** Quality and usefulness of ER diagrams, SQL demonstrations, and real-world examples.
- ***Audio/Video Quality:*** Clarity of audio, screen resolution, and legibility of on-screen content.
- ***Engagement Level:*** Instructor's ability to maintain learner interest throughout the session.

***Assignments and Challenges***

- ***Relevance to Module Content:*** Alignment with theoretical topics covered in each module.
- ***Difficulty Level:*** Balance between ease and challenge to promote critical thinking.
- ***Instruction Clarity:*** Clarity and specificity of problem statements.
- ***Feedback Quality:*** Usefulness and completeness of post-submission explanations.
- ***Time Required:*** Reasonableness of time needed to complete each challenge.

***Practical Sessions***

- ***Hands-onLearning Value:*** Effectiveness of SQL tasks in reinforcing lecture concepts.
- ***Tool Usability:*** Ease of using MySQL, Oracle, or online simulators for practice.
- ***Setup Ease:*** Difficulty in installing and configuring the required database environment.
- ***Theory Correlation:*** Degree to which practical tasks align with and reinforce video content.

### *3.4 Tools Used*

| *Tool* | *Purpose* |
|---|---|
| Google Forms | Survey design, distribution, and response collection |
| Microsoft Excel | Statistical analysis, chart generation, and mean computation |
| Likert Scale (1-5) | 1 = Poor/Strongly Disagree; 5 = Excellent/Strongly Agree |
| MySQL 8.0 | Execution of SQL practicals referenced in the course |
| DB Fiddle (online) | Browser-based SQL practice for students without local MySQL |
| Thematic Coding | Categorisation of open-ended qualitative responses into themes |

***Table 3.1: Tools Used for Data Collection and Analysis***

# CHAPTER 4
# *ANALYSIS AND FINDINGS*

This chapter presents a detailed analysis of the data collected through the peer survey and personal course evaluation. The findings are supported by statistical measures including mean scores, percentage distributions, and qualitative themes from student feedback.

## *4.1 Overall Course Performance*

The overall course performance was assessed by computing the mean rating across all three components -- video lectures, assignments, and practicals -- from 35 survey respondents.

***Overall Mean Rating: 4.2 / 5.0***

This rating indicates that the course is generally well-received and meets its primary objective of delivering DBMS knowledge effectively. A large proportion of students expressed satisfaction with the course content and module structure.

| *Rating* | *No. of Students* | *Percentage* |
|---|---|---|
| 5 (Excellent) | 14 | 40% |
| 4 (Very Good) | 5 | 14% |
| 3 (Good) | 2 | 6% |
| 2 (Average) | 0 | 0% |
| 1 (Poor) | | |

***Table 4.1: Overall Course Rating Distribution (n=35)***

The data shows that 80% of students rated the course as either 'Very Good' or 'Excellent', indicating strong overall performance. A small percentage reported dissatisfaction primarily due to practical tool-setup difficulties.

## *4.2 Video Lectures*

Video lectures were analysed across five parameters. They emerged as the strongest component of the course, driven by the instructor's ability to connect DBMS theory to

real-world industry scenarios and the quality of worked examples.

| *Parameter* | *Mean Score (/5)* | *Key Observation* |
|---|---|---|
| Clarity of Concepts | 4.6 | Complex joins and normalisation explained with clear analogies<br>Slightly fast in normalisation modules 8-9 |
| Pacing and Length | 4.3 | ER diagrams and live SQL demos were highly appreciated |
| Visual Aids (ER/SQL) | 4.5 | Occasional background noise in a few lectures |
| Audio/Video Quality | 4. | Industry examples kept attention high throughout |
| Engagement Level | 2 | -- |
| Overall Lectures | 4. | |

*Table 4.2: Mean Scores -- Video Lecture Parameters*

The SQL Joins and Subqueries module received the highest individual lecture score (4.7/5), while the Concurrency Control module scored lowest (4.0/5), with students citing abstract notation. The IMDB Case Study (Module 16) was praised by 82% of respondents as a unique value-add.

***Student Feedback (Selected Qualitative Comments):***

> ***"Concepts were explained very clearly, especially normalisation and SQL joins." "Lectures are well-structured but the concurrency module felt a bit rushed."***
>
> ***"The IMDB case study is exactly what we needed -- real data, real queries."***

### *4.3 Assignments*

The 16 module challenges were evaluated for relevance, difficulty balance, instruction clarity, and feedback quality. Overall mean: ***3.9 / 5.0***.

| *Difficulty Level* | *No. of Respondents* | *Percentage* |
|---|---|---|
| Too Easy | 18 | 52% |
| About Right | 10 | 29% |
| Moderately Hard | | |

| Difficulty Level | No. of Respondents | Percentage |
|---|---|---|
| Too Difficult | 3 | 8% |

*Table 4.3: Assignment Difficulty Distribution (n=35)*

| Parameter | Mean Score (/5) |
|---|---|
| Relevance to | 4.3 |
| Module Difficulty | 3.7 |
| Balance Instruction | 4.1 |
| Clarity Feedback | 3.2 |
| Quality Time Req | 4.0 |

*Table 4.4: Assignment Evaluation Scores*

Assignments were highly rated for relevance and instruction clarity. However, feedback quality (3.2/5) was the lowest-rated parameter -- the platform provides correctness indicators but not detailed model solutions, limiting learning from mistakes.

### 4.4 Practicals

Practical sessions received an overall mean of ***3.6 / 5.0***, the lowest of the three components. While students valued the applied SQL tasks, tool-setup barriers significantly impacted the experience.

| Parameter | Mean Score (/5) | Observation |
|---|---|---|
| Hands-on SQL Value | 3.5 | MySQL setup frustrating for beginners |
| Tool Usability | 3.0 | No guided installation tutorial in course |
| Setup Ease | 4.2 | Practical tasks reinforce lectures well |
| Theory Correlation | 3.6 | -- |
| Overall Practicals | 4.1 | IMDB exercises highly valuable |

*Table 4.5: Practical Session Evaluation*

| Technical Issue | Percentage of Students |
|---|---|
| MySQL Installation Problems | 40% |
| Configuration / Driver Errors | 30% |

| *Technical Issue* | *Percentage of Students* |
|---|---|
| No Issues Encountered | 30% |

***Table 4.6: Technical Issues Faced During Practicals (n=35)***

Despite these challenges, students who completed practicals reported improved understanding of SQL and relational schema design. 40% of students used browser-based simulators (DB Fiddle, SQLiteOnline) as workarounds.

### *4.5 Comparative Analysis*

| *Component* | *Mean Score (/5)* | *Rank* | *Strongest Parameter* | *Weakest Parameter* |
|---|---|---|---|---|
| Video Lectures | 4.4 | 1st | Concept Clarity (4.6) | A/V Quality (4.2) |
| Assignments | 3.9 | 2nd | Relevance (4.3) | Feedback Quality (3.2) |
| Practicals | 3.6 | 3rd | Theory Correlation (4.2) | Setup Ease (3.0) |

***Table 4.7: Component-wise Comparative Analysis***

The comparative analysis confirms that the course is strongest in theoretical delivery through video lectures, and weakest in practical accessibility. This pattern is consistent with many MOOCs that are optimised for conceptual instruction rather than hands-on lab-style learning.

# CHAPTER 5
# *DISCUSSION AND RECOMMENDATIONS*

This chapter interprets the findings from Chapter 4, compares the course with standard undergraduate DBMS syllabi, and provides actionable recommendations to improve its effectiveness.

## *5.1 Discussion: Strengths and Weaknesses*

### *Strengths*

Based on the analysis, the course demonstrates several notable strengths when evaluated against the standard B.Tech DBMS syllabus:

- ***Comprehensive Syllabus Coverage:*** The 16 modules collectively cover approximately 85-90% of a standard second-year B.Tech DBMS syllabus, including ER modelling, relational algebra, complete SQL, normalisation (1NF to BCNF), transactions, concurrency control, indexing, and query optimisation.
- ***High-Quality Video Lectures:*** The instructor's industry background enables him to
contextualise abstract database concepts within real-world system design -- a dimension rarely present in textbook-driven classroom teaching.
- ***Real-World IMDB Case Study:*** Module 16's applied SQL project on the IMDB dataset is a standout feature, offering the kind of project-based learning that bridges theory and practice.
- ***Free Access with Certificate:*** Zero cost and a shareable Certificate of Excellence
significantly lower the barrier to entry, making the course accessible to all undergraduate students.
- ***Structured Module Progression:*** The course follows a logical learning arc from data modelling to implementation to advanced optimisation, aligning well with academic curricula.
- ***Self-paced with Lifetime Access:*** Students can revisit lectures as many times as needed, which is especially useful for revision before university examinations.

• ***Absent Model Solutions:*** The platform provides challenge correctness feedback but no detailed worked solutions, making it difficult for students to learn from their errors.

• ***No Guided Tool Setup:*** The course recommends MySQL but provides no embedded installation tutorial, leading to setup failures for 40% of students.

• ***Limited Forum Moderation:*** Discussion threads receive infrequent instructor or moderator responses, leaving complex doubts unresolved.

• ***Superficial NoSQL Coverage:*** Module 15 covers only MongoDB basics without sufficient depth for students pursuing distributed systems or cloud computing electives.

• ***One-way Communication:*** Video lectures are pre-recorded and non-interactive, reducing engagement compared to real-time classroom instruction for complex topics.

## *5.2 Recommendations*

### *Improvement in Video Lectures*

Although video lectures are the strongest component, further improvements can be made to increase engagement and accessibility:

• Incorporate animated ER-to-relational schema conversion walkthroughs for Modules 2 and 3, where schema derivation is a frequently examined topic.
• Break longer lectures (exceeding 25 minutes) into shorter segments of 15-20 minutes each to improve concentration and retention.
• Add in-video quiz checkpoints at key concept transitions to promote active learning.
• Include a dedicated sub-lesson on Concurrency Control with visual timeline diagrams to address the lowest-rated lecture topic.

### *Enhancement of Assignments*

Assignments are well-aligned but require improvements in feedback and difficulty calibration:

• Publish detailed model answers with step-by-step explanations after the submission deadline for each challenge.
• Introduce auto-graded SQL coding exercises (not only MCQ) to provide hands-on practice within the platform.

• Redesign challenges to include a mix of basic, intermediate, and advanced questions, ensuring consistent difficulty distribution.

• Add a comprehensive end-of-course SQL project with peer review to encourage collaborative assessment.

***Strengthening Practical Sessions***

Practical learning requires the most significant improvement, primarily by reducing technical access barriers:

• Integrate an embedded cloud-based SQL sandbox (e.g., DB Fiddle) directly into the course interface, eliminating the need for local MySQL installation.

• Create a dedicated 10-15 minute video tutorial for MySQL/PostgreSQL setup and configuration, placed at the beginning of Module 5 (SQL-DDL).

• Expand practical tasks to include real-world scenarios such as designing schemas for e-commerce systems, library management, and student portals.

• Provide a troubleshooting FAQ document for common MySQL installation and driver configuration errors.

| *Recommendation* | *Component* | *Priority* |
|---|---|---|
| Publish detailed model solutions after each challenge | Assignments | High |
| Embed cloud-based SQL sandbox in the platform | Practicals | High |
| Add animated ER schema derivation walkthroughs | Lectures | Medium |
| Introduce auto-graded SQL coding exercises | Assignments | High |
| Add MySQL setup video tutorial at start of Module 5 | Practicals | Medium |
| Break 25+ minute lectures into shorter segments | Lectures | Medium |
| Enable weekly instructor-moderated Q&A; threads | Platform | Medium |

| *Recommendation* | *Component* | *Priority* |
|---|---|---|
| Expand NoSQL module with Cassandra/Redis use-cases | Lectures | Low |
| Add visual concurrency timeline simulator | Lectures | Low |
| Add end-of-course SQL project with peer review | Assignments | High |

***Table 5.1: Recommendations Summary with Priority***

# CHAPTER 6
# *CONCLUSION*

This project evaluated the 30-hour online course on Database Management Systems offered by Scaler Topics, examining its effectiveness across three core dimensions: video lectures, assignments and challenges, and practical sessions. The study combined personal course completion with a structured Likert-scale survey administered to 35 second-year B.Tech (CSE/IT) peers.

The findings confirm that the course is a high-quality learning resource for undergraduate DBMS students. Video lectures scored the highest (4.4/5), with exceptional ratings for conceptual clarity and visual aids. The instructor's industry experience translates into clear, contextualised explanations of complex topics such as SQL joins, normalisation, ACID properties, and query optimisation. The IMDB dataset case study (Module 16) stands out as a particularly impactful module that connects theory to real-world application.

Assignments and challenges are well-aligned with module content (3.9/5) but are limited by the absence of detailed post-submission solutions. This gap reduces the learning value of incorrect attempts, which is a critical drawback compared to traditional academic assessment practices.

Practical sessions, while conceptually valuable (Theory Correlation: 4.2/5), received the lowest component score (3.6/5) due to tool-setup barriers. The absence of a guided MySQL installation tutorial led to setup failures for 40% of survey respondents, significantly limiting the practical learning experience.

Overall, the course is well-suited for second-year B.Tech students as a ***supplementary learning resource***, particularly for concept revision, placement preparation, and SQL skill-building. It covers approximately 85-90% of a standard undergraduate DBMS syllabus with commendable instructional quality. However, as a standalone replacement for university lab instruction, it currently falls short in assessment depth and practical accessibility.

With the implementation of the recommended improvements -- embedded SQL sandboxes, detailed model solutions, animated walkthroughs, and instructor-led Q&A; forums -- this course has the potential to become a comprehensive, institution-grade MOOC for undergraduate Database Management Systems education.

***Future Scope***

- A longitudinal study comparing academic performance of students who used this course against those who did not would provide stronger evidence of its educational impact.
- A multi-platform comparison across DBMS MOOCs (NPTEL, Coursera, Scaler)
would enable benchmarking and selection guidance for institutions.
- Extension of the survey sample to multiple institutions across different states would
improve the statistical generalisability of findings.
- Evaluation from the instructor's perspective -- covering pedagogical design decisions and platform constraints -- would provide a complete picture of MOOC effectiveness.

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